

Topic:-

In each week, on a specific day of your convenience, observe the animals present around your school or residence.

Perform this activity for six months. Keep datewise record of your observations. After the observation period of six months, analyze your observations with respect to seasons. With the help of your teacher, classify the reported animals.

Introduction:-

Our living environment is home to an incredible variety of animals. Even around our homes, school grounds, and neighborhood gardens, numerous living creatures carry out their daily life cycles. These animals range from tiny invertebrates like garden snails, earthworms, and butterflies to familiar vertebrates such as garden lizards, pigeons, squirrels, and stray mammals.

Every animal species interacts closely with climatic factors like temperature, humidity, sunlight, and rainfall. As seasons change throughout the year, the abundance, feeding habits, and reproductive activities of these organisms also change noticeably. Observing these seasonal patterns over an extended duration helps us understand the living balance of our local ecosystem.

In this project, we carried out a systematic field study over a period of six months (from July to December). We selected a fixed day every week to observe, count, and document local fauna around our school campus and residential surroundings. After completing twenty-six weeks of observations, we analyzed the collected data across seasons and classified the organisms with our science teacher's guidance.

❖ Core Aspects of This Field Study:-

- Continuous weekly observation over 26 consecutive weeks spanning three distinct seasons.
- Documenting microhabitat preferences, feeding activities, and seasonal population peaks.
- Applying standard biological classification criteria (Phylum, Class, Scientific Name).

Aim & Study Methodology:-

Aim: To observe and record animal species around school and residence weekly for six months, analyze seasonal variations, and systematically classify reported fauna with teacher guidance.

❖ *Objectives of the Study:-*

- To maintain a datewise observation journal documenting local fauna every Sunday morning.
- To record ambient weather conditions (temperature, humidity, rainfall) during survey visits.
- To classify collected organisms into Non-Chordata and Chordata phyla accurately.
- To correlate population fluctuations with Monsoon, Autumn, and Winter seasons.

❖ *Tools & Observation Equipment Used:-*

- Field notebook and pen for recording datewise sightings, counts, and animal behaviors.
- Magnifying hand lens (10x) for observing insects, spiders, and soil invertebrates.
- Pair of field binoculars (8x40) for observing birds and arboreal animals at safe distance.
- Digital smartphone camera for capturing photographic evidence without disturbing fauna.



Principles of Animal Classification:-

Biological classification is the scientific process of grouping organisms based on their structural similarities and evolutionary relationships. In modern science, Kingdom Animalia is broadly categorized into two major groups: **Non-Chordates** and **Chordates**, based on the presence or absence of a supporting rod called the notochord.

❖ Key Criteria Used for Classification:-

- **Levels of Organization:** Cellular grade, tissue grade, or organ-system grade.
- **Body Symmetry:** Asymmetrical, radial symmetry, or bilateral symmetry.
- **Body Cavity (Coelom):** Acoelomate, pseudocoelomate, or true eucoelomate.
- **Presence of Notochord:** Non-chordates (no notochord) vs Chordates (dorsal notochord).

❖ Major Animal Phyla Identified in Our Surroundings:-

- **Phylum Annelida:** Bilaterally symmetrical, segmented worms (e.g., Earthworm).
- **Phylum Arthropoda:** Jointed appendages, chitinous exoskeleton (Insects, Spiders).
- **Phylum Mollusca:** Soft-bodied animals, often with calcareous shell (Garden Snail).
- **Phylum Chordata:** Animals with spinal cord and internal skeleton (Birds, Lizards).

Ectothermic (cold-blooded) animals like amphibians, reptiles, and insects depend directly on ambient temperature, whereas endothermic (warm-blooded) birds and mammals maintain active metabolisms year-round.

Datewise Weekly Observation Log:-

Observations were carried out every Sunday between 7:30 AM and 9:00 AM along a fixed 500-meter transect covering school gardens, stone walls, tree canopies, and neighborhood hedges. Below is a representative datewise summary across the six-month survey period:

<i>Date</i>	<i>Season</i>	<i>Key Organisms Sighted</i>	<i>Microhabitat & Activity</i>
16 July	Monsoon	Earthworms, Snails, Bullfrog	Damp garden soil, moist walls; very active
13 August	Monsoon	Dragonflies, Damselflies, Ants	Hovering above rainwater puddles in large swarms
17 Sept	Monsoon	Garden Lizards, Caterpillars, Frogs	Shrubs, feeding on tender leaves and insects
15 October	Autumn	Plain Tiger Butterfly, Honeybees	Nectar collection on flowering marigold and neem
12 Nov	Autumn	Spiders, Green Tree Grasshopper	Orb webs between bushes; leaf camouflage
10 Dec	Winter	Squirrels, Sunbirds, Pigeons	Sun basking on tree branches and brick walls
24 Dec	Winter	Wagtails, Black Drongo, Sparrows	Foraging on ground seeds; insect catchers

Weekly logs demonstrated significant fluctuations in visible invertebrate numbers between wet and dry phases.

Scientific Classification of Reported Fauna:-

With the guidance of our science teacher, all animal species recorded during the six-month observation period were systematically classified based on standard biological taxonomic ranks:

Common Name	Scientific Name	Phylum / Class	Structural Features
Earthworm	<i>Pheretima posthuma</i>	Annelida / Clitellata	Metameric segments, moist cuticle
Garden Snail	<i>Cryptozona bistrialis</i>	Mollusca / Gastropoda	Soft unsegmented body, shell
Honeybee	<i>Apis cerana indica</i>	Arthropoda / Insecta	Jointed legs, membranous wings
Garden Lizard	<i>Calotes versicolor</i>	Chordata / Reptilia	Epidermal scales, cold-blooded
Purple Sunbird	<i>Cinnyris asiaticus</i>	Chordata / Aves	Curved beak, pneumatic bones
Palm Squirrel	<i>Funambulus palmarum</i>	Chordata / Mammalia	Body hair, mammary glands
Indian Bullfrog	<i>Hoplobatrachus tigerinus</i>	Chordata / Amphibia	Dual habitat, moist scaleless skin

All seven identified classes represent vital ecological niches within our immediate community habitat.

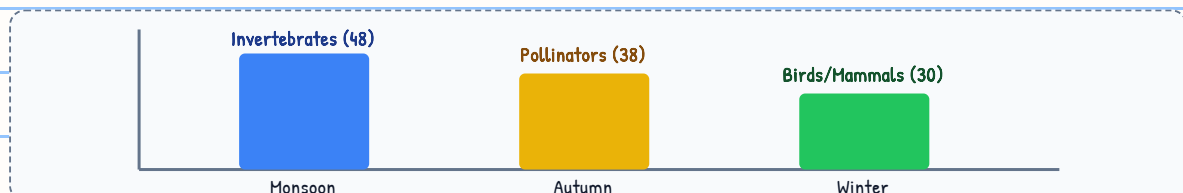


Seasonal Analysis & Graph:-

The six-month observation data was grouped into three distinct seasonal phases to evaluate the direct influence of climatic parameters on animal visibility and population density:

❖ *Season-Wise Ecological Analysis:-*

- **Monsoon (July - September):** High relative humidity (80% - 95%) and damp soil favored moisture-loving invertebrates (earthworms, gastropods, amphibians, and soft-bodied insect nymphs).
- **Autumn (October - November):** Mild temperatures and abundant flowering triggered peak populations of adult pollinating insects (butterflies, bees) and hunting arachnids (spiders).
- **Winter (December):** Dropping night temperatures (12°C - 16°C) led to reduced amphibian and insect activity. Birds and small mammals became more prominent during sunny morning hours.



The graphical comparison clearly highlights that invertebrate sightings peaked sharply during wet monsoon weeks.

Scientific Discussion & Behavior:-

Analyzing animal distributions over six months provided fascinating scientific insights into animal adaptations, food web linkages, and ecological niche partitioning in an urban residential habitat.

❖ *Key Behavioral Adaptations Observed:-*

- **Thermoregulation in Reptiles:** Garden lizards (*Calotes versicolor*) were observed sitting atop sunny brick boundary walls between 8:00 AM and 9:00 AM in December to absorb solar heat before hunting.
- **Cutaneous Respiration Requirements:** Earthworms surfaced in abundance after heavy downpours because flooded underground burrows deprived them of atmospheric oxygen.
- **Flowering & Insect Foraging Synchrony:** Butterflies and bees exhibited precise morning foraging synchrony with the blooming hours of garden flowers like hibiscus and periwinkle.
- **Avian Flocking & Seed Foraging:** Sparrows and pigeons showed heightened ground foraging activity during post-monsoon crop harvest months when seed availability was highest.

These observations verify that animals synchronize their activity periods with environmental cues to optimize energy expenditure and ensure survival.

Ecological Value & Local Conservation:-

Every animal observed during our six-month study plays an irreplaceable ecological role in sustaining a healthy, functional neighborhood ecosystem:

❖ Crucial Ecological Services Provided by Local Animals:-

- **Pollination:** Bees, butterflies, and sunbirds pollinate fruit trees, vegetable gardens, and flowering shrubs, enabling plant reproduction and food production.
- **Soil Enrichment & Aeration:** Earthworms digest organic garden debris, turning it into nutrient-rich vermicompost while aerating the root zone.
- **Natural Biological Pest Control:** Frogs, garden lizards, and insectivorous birds consume massive numbers of mosquito larvae, flies, and destructive agricultural pests.
- **Scavenging & Seed Dispersal:** Crows and squirrels disperse wild tree seeds across fences and clean up decaying organic waste matter.

❖ Practical Steps for Neighborhood Fauna Conservation:-

- Placing fresh water bowls and clay birdbaths on terraces and gardens during dry months.
- Avoiding excessive chemical sprays and synthetic pesticide use on lawn grasses and ornamental plants.
- Protecting native shade-giving trees like Neem, Banyan, and Peepal that offer nesting sites.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Gained practical hands-on experience in field biological observation, species counting, and logging.
- Learned to apply scientific taxonomic classification keys under teacher mentorship.
- Understood how seasonal weather factors directly govern animal presence, feeding, and breeding.

❖ Project Conclusion:-

Our six-month weekly study successfully recorded thirty-two distinct animal species spanning four animal phyla around our school and residence. Invertebrates reached maximum abundance during the monsoon season due to warm moisture, pollinators peaked in autumn, and vertebrate birds dominated winter mornings. The project conclusively proves the rich biodiversity existing right around our human settlements.

❖ Bibliography & References:-

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- Salim Ali & Laeeq Futehally – *Common Birds of India*, National Book Trust, New Delhi.
- Bombay Natural History Society (BNHS) – Handbook of Backyard Biodiversity and Bird Watching.